$^4C_4 \quad u^0 \quad v^4$

General Formula: $f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$

a) $f(x) = \ln(x) \qquad a = 1$

when $x = a$

$$f(x) = \ln(x) \qquad\qquad = 0$$

$$f'(x) = \frac{1}{x}, \quad x^{-1} \qquad = 1$$

$$f''(x) = \frac{-1}{x^2}, \quad -x^{-2} \qquad = -1$$

$$f'''(x) = \frac{2}{x^3}, \quad 2x^{-3} \qquad = 2$$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$= 0 + \frac{1(x-1)}{1!} + \frac{-1(x-1)^2}{2} + \frac{\cancel{2}(x-1)^3}{3 \times \cancel{2}}$$

$$= +(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{n}$$

$\theta)$ $f(x) = e^{x}$ $a = 3$

when $x = a$

$f(x) = e^{x} = e^{3}$

$f'(x) = e^{x} = e^{3}$

$f''(x) = e^{x} = e^{3}$

$f'''(x) = e^{x} = e^{3}$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$= e^3 + \frac{e^3(x-3)}{1!} + \frac{e^3(x-3)^2}{2!} + \frac{e^3(x-3)^3}{3!}$$

$$= \sum_{n=0}^{\infty} \frac{e^3(x-3)^n}{n!}$$

Q $\sin(x)$ $\quad x = \frac{\pi}{3}$ Dont Do this

when $x=a$

$$f(x) = \sin x = \frac{\sqrt{3}}{2}$$

$$f'(x) = \cos x = \frac{1}{2}$$

$$f''(x) = -\sin x = -\frac{\sqrt{3}}{2}$$

$$f'''(x) = -\cos x = -\frac{1}{2}$$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$\frac{\sqrt{3}}{2} + \frac{\frac{1}{2}\left(x-\frac{\pi}{3}\right)}{1!} \quad \frac{-\frac{\sqrt{3}}{\cancel{2}}\left(x-\frac{\pi}{3}\right)^2}{\cancel{2!}} - \frac{\frac{1}{2}\left(x-\frac{\pi}{3}\right)^3}{3!}$$

$$\frac{\sqrt{3}}{2} + \frac{1}{2}(58.95) - \sqrt{3}(58.95)^2 \quad \frac{1}{3}(58.95)^3$$

$$+ 29.476 - \sqrt{3}(3475.4325$$